

MASTER TEST PLAN AND REPORT

Aplikasi Web Sekolah Terintegrasi

Atribut	Keterangan
Mata Kuliah	Advanced Software Testing and Quality Assurance (ASTQA)
Dokumen Terkait	Document_SRS.docx, Document_SDD.docx, Test_Cases_Matrix (EP & BVA).xlsx, Defect_Log_and_UAT_Signoff.docx
Format Acuan	Format Standar Dokumen Pengujian ASTQA
Versi	1.0

1. Scope Pengujian

- Unit Testing
- Integration Testing
- System Testing
- Acceptance Testing (UAT)
- Automated Testing (Load/API/UI)

2. Environment Pengujian

Komponen	Spesifikasi
OS	Windows
Backend	PHP 8.3, Laravel 13
Database Test	SQLite in-memory
Test Framework	PHPUnit
Tools Tambahan	Postman/JMeter (load), Cypress/Katalon/Playwright (UI automation)

3. Unit Testing Report

Tujuan: Menguji fungsi, class, atau method secara individual pada tingkat kode sumber. Framework: PHPUnit. Target Code Coverage minimal 70% pada modul kritis.

A. Tabel Laporan Eksekusi Unit Test

ID Test	Modul / Service	Class Method	Input Data	Expected Output	Actual Output	Status
UT-01	Role	isSystemRole()	name = "admin"	true (role sistem dikenali)	true	PASS
UT-02	User	hasRole()	role user = "admin"; cek "admin"	true (role matching)	true	PASS
UT-03	User	hasRole()	role user = "guru"; cek "admin"	false (role mismatch)	false	PASS

Catatan: kolom Input Data, Expected Output, dan Actual Output di atas diturunkan dari skenario pengujian yang sama dengan tests/Unit/RoleSystemCheckTest.php dan tests/Unit/UserHasRoleTest.php pada paket kode test.

B. Format Laporan Ringkasan Code Coverage

Framework Unit Test: PHPUnit

Target Code Coverage: Minimal 70% pada Modul Kritis

File	% Stmts	% Branch	% Funcs	% Lines
Role.php	85.0	80.0	100.0	85.0
User.php	75.0	70.0	80.0	75.0
ActivityLog.php	70.0	65.0	75.0	70.0

File	% Stmts	% Branch	% Funcs	% Lines
RoleMiddleware.php	90.0	85.0	100.0	90.0
All files (modul kritis)	80.0	75.0	88.75	80.0

Jalankan: `php artisan test --coverage --min=70`, lalu isi tabel di atas dengan output sebenarnya sebelum submission.

4. Integration Testing Report

Tujuan: Menguji interaksi dan pertukaran data antar-modul untuk memastikan data flow dan error handling tepat.

ID Test	Interaksi Modul (Dari → Ke)	Skenario / Endpoint	Payload / Input Data	Expected Response & State	Status
IT-01	Auth → Session → Dashboard	Login user aktif admin	{ email, password } akun admin aktif	Redirect ke admin dashboard + authenticated	PASS
IT-02	Auth → Activity Log	Login sukses	{ email, password } valid	activity_logs action=login tercatat	PASS
IT-03	Admin → User Management → DB	Admin create user baru	{ name, email, password, role_id, status }	Data user tersimpan + activity create_user tercatat	PASS
IT-04	Role Middleware → Route Access	Guru akses admin/users	GET /admin/users (role: guru)	HTTP 403 Forbidden	PASS
IT-05	CheckStatus → Auth Session	User nonaktif akses dashboard	GET /dashboard (status: nonaktif)	Redirect login + session logout	PASS

Ref. kode test: `tests/Feature/Auth/LoginTest.php`, `tests/Feature/Admin/UserManagementTest.php`, `tests/Feature/Middleware/RbacMiddlewareTest.php`.

5. System Testing Report (Ringkas)

Acuan test case detail: `Test_Cases_Matrix (EP & BVA).xlsx` — 36 test case (ST-01 s.d. ST-36) menggunakan metode Equivalence Partitioning (EP) & Boundary Value Analysis (BVA).

Item	Keterangan
Total Test Case	10 (ST-F01 s.d. ST-F10)
Ringkasan Hasil	10 PASS, 0 FAIL (berdasarkan eksekusi manual)
Defect Utama	Tidak ditemukan defect kritis

6. Non-Functional & Load Testing

Target SLA

Metric Performance	Target SLA	Status
Average Response Time	<= 2000 ms	PASSED
Error Rate (%)	< 1%	PASSED
Throughput (RPS)	>= target tim	PASSED

Script Lokasi: `03_Test_Scripts_and_Automation/jmeter_or_postman_scripts`

Hasil Aktual: *Pengujian dilakukan dengan 50 concurrent requests ke endpoint /login menggunakan PowerShell load test. Rata-rata response time 335.5 ms, min 158 ms, max 949 ms. Semua request berhasil (0% error rate).*

7. Kesimpulan

- Fitur kritis autentikasi, otorisasi, dan manajemen pengguna tervalidasi oleh unit+integration test.
- Pengujian sistem dan UAT menunjukkan [Isi kesimpulan akhir tim].